

# Qingzhao Zhang

Assistant Professor, Electrical and Computer Engineering (ECE), University of Arizona

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## RESEARCH INTERESTS

System security (e.g., cyber-physical systems, autonomous driving), software security (e.g., program analysis, formal verification), AI security (e.g., adversarial attacks, robustness).

## EDUCATION

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| <b>University of Michigan</b><br><i>Ph.D. in Computer Science and Engineering</i> | <i>Sep 2019 – May 2025</i> |
| <b>University of Michigan</b><br><i>M.S. in Computer Science and Engineering</i>  | <i>Sep 2019 – Dec 2023</i> |
| <b>Shanghai Jiao Tong University</b><br><i>B.E. in Computer Engineering</i>       | <i>Sep 2015 – May 2019</i> |

## WORK EXPERIENCE

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| <b>Assistant Professor   University of Arizona, ECE</b><br>• Lead <i>Secure Autonomous Systems</i> (SAS) research group working on computer security and systems. We advance the security, resilience, and trustworthiness of emerging autonomous systems (e.g., self-driving vehicles, drones, and robots) through cross-layer design and analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <i>Aug 2025 - Present</i>   |
| <b>Research Assistant/Research Fellow   University of Michigan, EECS</b><br>Supervisor: Prof. Z. Morley Mao<br>• Research on software/system for cyber-physical system safety — <i>AVChecker</i> , the first traffic rule compliance checker on autonomous driving software; <i>SmtConf</i> , safety vetting of industrial control system configuration.<br>• Research on adversarial machine learning on cyber-physical systems — Adversarial attack and mitigation on trajectory prediction on autonomous driving; Analyzed data fabrication vulnerability on collaborative perception.<br>• Research on robustness of vehicular network — <i>RAO</i> , collaborative perception with asynchronous sensors.<br>• Research on robust perception algorithms of autonomous driving, and large language model security/efficiency. | <i>May 2020 - Aug 2025</i>  |
| <b>Software Engineer Intern   Google</b><br>• Designed and implemented formal verification solutions to enhance the security properties of an embedded system kernel (based on open-sourced Tock OS), involving modular model checking and theorem proving.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>May 2023 - July 2023</i> |
| <b>Software Engineer Intern   Google</b><br>• Designed and implemented static analysis checks based on Android Lint for Google's Android tests, which was deployed to assist Google developers to write high-quality unit tests.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <i>May 2022 - July 2022</i> |

## SELECTED PUBLICATIONS

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| <b>Conference Papers</b><br>• <b>[S&amp;P '26]</b> Jiarui Li, Joey Brewington, <b>Qingzhao Zhang</b> , Z. Morley Mao. "Banshee: Target Switch Attacks on Gimbal-Stabilized Visual Tracking Systems via Acoustic Injection", <i>The 47th IEEE Symposium on Security and Privacy</i> .<br>• <b>[COLM '25]</b> Shuwei Jin, Yongji Wu, Haizhong Zheng, <b>Qingzhao Zhang</b> , Matthew Lentz, Z Morley Mao, Atul Prakash, Feng Qian, Danyang Zhuo. "Plato: Plan to Efficiently Decode for Large Language Model Inference", <i>Conference on Language Modeling</i> . |
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- [ICML '25] Shuowei Jin\*, Xueshen Liu\*, **Qingzhao Zhang**, Z. Morley Mao. “Compute Or Load KV Cache? Why Not Both?”, *Forty-Second International Conference on Machine Learning*.
- [ICLR '25] Minkyong Cho, Yulong Cao, Jiachen Sun, **Qingzhao Zhang**, Marco Pavone, Jeong Joon Park, Heng Yang, Z. Morley Mao. “Cocoon: Robust Multi-Modal Perception with Uncertainty-Aware Sensor Fusion”, *The International Conference on Learning Representations*.
- [USENIX Security '24] **Qingzhao Zhang**, Shuowei Jin, Ruiyang Zhu, Jiachen Sun, Xumiao Zhang, Qi Alfred Chen, Z. Morley Mao. “On Data Fabrication in Collaborative Vehicular Perception: Attacks and Countermeasures”, *Proceedings of the 33rd USENIX Security Symposium*.
- [ICLR '24] Jiachen Sun, Haizhong Zheng, **Qingzhao Zhang**, Atul Prakash, Z. Morley Mao, Chaowei Xiao. “CALICO: Self-Supervised Camera-LiDAR Contrastive Pre-training for BEV Perception”, *The International Conference on Learning Representations*.
- [MobiCom '23] **Qingzhao Zhang\***, Xumiao Zhang\*, Ruiyang Zhu\*, Fan Bai, Mohammad Naserian, Z. Morley Mao. “Robust Real-time Multi-vehicle Collaboration on Asynchronous Sensors”, *Proceedings of the 29th Annual International Conference on Mobile Computing and Networking*.
- [RAID '22] **Qingzhao Zhang**, Xiao Zhu, Mu Zhang, Z. Morley Mao. “Automated Runtime Mitigation for Misconfiguration Vulnerabilities in Industrial Control Systems”, *Proceedings of the 25th International Symposium on Research in Attacks, Intrusions and Defenses*.
- [CVPR '22] **Qingzhao Zhang**, Shengtuo Hu, Jiachen Sun, Qi Alfred Chen, Z. Morley Mao. “On adversarial robustness of trajectory prediction for autonomous vehicles”, *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- [SIGMETRICS '21] **Qingzhao Zhang**, David Ke Hong, Ze Zhang, Qi Alfred Chen, Scott Mahlke, Z Morley Mao. “A systematic framework to identify violations of scenario-dependent driving rules in autonomous vehicle software”, *Proceedings of the ACM on Measurement and Analysis of Computing Systems*.

## TEACHING AND MENTORSHIP

### Course Instructor, University of Arizona

- Spring 2026, ECE 402C/502C Operating System Design.

### PhD Students, University of Arizona

- Yutong Liu, started in Fall 2025.

### Supervisor of Undergraduate Research, University of Michigan

2022

- Multidisciplinary Design Program (MDP) at University of Michigan. Lead cybersecurity research and supervised a team of four students.

### Research Mentorship, University of Michigan

2020-2024

- Research on program analysis, autonomous driving, adversarial machine learning.
- Mentees: Charles Ziegenbein Jr., Kevin Zhang, Andrew Wei, Xingyu Wang, Jingjia Peng, Ziyang Xiong, Runting Zhang.

## SERVICES

- Conference Committee Member: S&P 2027, AsiaCCS 2027, CCS 2026/2025
- Conference Reviewer: NeurIPS 2026, COLM 2026, CVPR 2026, ECCV 2026, ICLR 2026/2025, ICRA 2025, ACM MM 2024, IV 2024/2023
- Journal Reviewer: TDSC, IoT-J, Sensors, TIFS, ITSM
- Workshop Committee Member: RICSS 2024, RICSS 2025
- Artifact Reviewer: Usenix Security 2022
- Pre-review Task Force: NSDI 2025